

What Timing and Behavior Patterns Determine Speed Dating Success in Japan?

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Abstract

Choosing a life partner is a significant decision requiring understanding preferences and compatibility. Speed dating provides a structured setting for exploring these aspects through direct interactions within limited time. Understanding specific behaviors during speed dating that influence outcomes could benefit participants and service providers significantly. However, the precise timing and behavior patterns affecting speed dating outcomes remain underexplored. This paper reports exploratory experiments analyzing multimodal features over time using machine learning to uncover when and which behaviors impact speed dating success. The results suggest multimodal information from 2–4 and 9–10 minute segments predicts outcomes as accurately as data from the entire session. Additionally, linguistic features play a particularly significant role among multimodal features.

CCS Concepts

• **Human-centered computing** → **HCI theory, concepts and models**; *Collaborative interaction*; *Collaborative and social computing theory, concepts and paradigms*.

Keywords

Speed dating, Multimodal interaction, Machine learning

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1 Introduction

Choosing a life partner is a significant decision that requires understanding preferences and compatibility. Speed dating supports this process by providing a structured setting where individuals can explore these aspects within a limited time through direct interactions [9]. Understanding the specific behaviors during speed dating that influence matching outcomes could bring significant benefits to both participants and service providers. For participants, adopting behaviors that increase the likelihood of success and avoiding those that lead to failure can enhance their chances of finding a partner. For service providers, it enables them to offer tailored behavioral guidelines to participants and facilitate opportunities between individuals with compatible interaction styles. However, the precise timing and patterns of behaviors that impact matching outcomes during speed dating remain underexplored. Approaches that rely on static attributes [9] or personality traits and values assessed through pre-speed dating questionnaires [2, 15, 16, 22, 28] fail to account for behaviors exhibited during communication. While post-event questionnaires [2, 4, 10, 12, 13, 20, 21, 27] may provide insights into impactful behaviors, they heavily depend on participants' subjective responses, which limits their accuracy, comprehensiveness, and objectivity. Analyzing observed information recorded through cameras and microphones could lead to the discovery of behaviors that influence outcomes. However, existing studies [17, 23, 24, 29] have conducted comprehensive analyses of entire speed dating sessions using only one or two features, such as acoustic or lexical features, making it difficult to understand which behavioral patterns at specific moments affected the results. While Ali et al. [1] performed a time-segmented analysis, their approach relied solely on facial features.

To identify which patterns of behavior at specific moments influence speed dating outcomes, we take a novel approach by analyzing multimodal information for each time segment, unlike previous studies. This paper serves as a prequel to this effort, reporting exploratory experiments that analyze visual, acoustic, and linguistic features across time segments using machine learning techniques

to uncover when and which patterns of behavior impacted the success of speed dating. The comprehensive workflow in this study is shown in Figure 1.

2 Related work

Finkel et al. [7, 8] highlight that speed dating procedures allow researchers to investigate romantic dynamics within dyadic interactions while maintaining strong external validity, offering valuable guidelines for collecting data in settings that closely resemble real-world contexts. While some studies infer speed dating outcomes based on attributes like race or region of origin [9], most research groups focus on predicting these outcomes using self-reported preferences and observed behaviors.

2.1 Self-report-based prediction

In the fields of social psychology and behavioral science, a traditional approach has been to predict speed dating outcomes based on participants' self-reported personality traits, preferences, and evaluations of their partners.

In most speed dating studies, romantic preferences are pre-measured to analyze their relationship with outcomes. Valentine et al. [28] demonstrated that individuals' stated preferences for warmth and trustworthiness reliably predicted their attraction to potential partners in a speed dating context. However, Joel et al. [16] found that predicting speed dating outcomes using random forests is challenging when relying solely on pre-measured romantic preferences. Pre-measured personality traits are known to play a crucial role in predicting speed dating results. McClure et al. [22] demonstrated that chronic attachment anxiety predicted the outcomes of speed dating interactions, highlighting its significant impact on factors such as selectivity, popularity, and overall matching success. Asendorph et al. [2] found that for women, a man's openness to experience and shyness significantly influences their decisions. Jauk et al. [15] showed that narcissism positively predicted mate appeal for both short-term and long-term relationships in a speed dating context, highlighting its role in relationship success.

Predictions based on metrics measured after speed dating sessions have also garnered increasing attention. Many studies report that physical attractiveness perceived during speed dating serves as a strong predictor of outcomes for both men and women [2, 10, 12, 13, 20]. Similarities in personalities and attitudes have also been reported to predict mutual interest and positive outcomes in speed dating interactions [27, 30]. Interestingly, Tidwell et al. [27] argue that it is not actual similarity but perceived similarity that significantly contributes to predicting initial attraction. Perceived behavior, as experienced by the speed dating participants themselves, is also an important predictor. Houser et al. [12] discovered that in brief 30-second speed dating interactions, positive behaviors perceived in a partner serve as key predictors of judgment for both men and women. The same research group [13] also revealed that during six-minute speed dating sessions, positive perceptions of partner's nonverbal behaviors influence the desire to further develop the relationship.

The influence of self-reported information is also known to vary depending on the design and context of the speed dating setting. Luo et al. [21] compared the results of a speed dating study involving

54 male-female pairs found that strong reciprocal liking occurs only when individuals are aware of others' feelings toward them. Bhargava et al. [4] experimentally demonstrated the existence of a contrast effect in a speed dating setting, where perceiving high attractiveness in the immediately preceding partners reduces the perceived attractiveness of the current partner.

2.2 Observation-based prediction

There are also studies that attempt to predict speed dating outcomes using objectively observed information rather than participants' self-evaluations.

Many discourse analyses [18, 19, 26] have revealed that in speed dating interactions, individuals strategically craft their statements to enhance intimacy and foster deeper connections. Jurafsky, Ranganath, et al. [17, 24] analyzed lexical and prosodic features extracted from a four-minute speed dating spoken corpus using machine learning techniques to predict whether a speaker is flirtatious, friendly, awkward, or assertive. The same research group [23] has attempted to detect a speaker's intent to flirt during speed dating interactions using observed lexical and prosodic features. Veenstra et al. [29] developed a machine learning model using visual features related to position, proximity, and motion to predict whether speed dating participants wanted to exchange contact information with their partner or found them physically attractive. Ali et al. [1] reported that analyzing the first or last segment of a conversation, rather than the whole conversation, allows for more accurate prediction of conversational skills during speed dating based on features related to facial action units.

3 Copus Data

In this study, we utilized the MMSD (Multi-Modal Speed Dating) corpus [14], which comprises recordings of 625 speed dating sessions involving 146 Japanese participants paired as men and women. The participants' ages ranged from their 20s to 60s, with an average age of 31.9 years and a standard deviation of 8.6 years. For this study, we used video data, audio data, transcribed textual data of spoken content, and matching outcome data from 10-minute speed dating sessions in the corpus. The video data captures each participant from the chest up, recorded from a slightly elevated, frontal angle. The audio data was collected using headset microphones worn by the participants. The matching outcome data includes participants' willingness to exchange contact information with their partners (e.g., "want to exchange contact information" or "do not want to exchange contact information").

4 Prediction model

We attempted to predict the outcomes of speed dating sessions based on participants' behaviors. Specifically, we extracted visual, acoustic, and linguistic features related to participants' behaviors during the sessions from the video data, audio data, and transcribed textual data in the MMSD corpus. We developed machine learning models to predict the outcome of speed dating using three features. Furthermore, we calculated the feature importance for the constructed machine learning model.

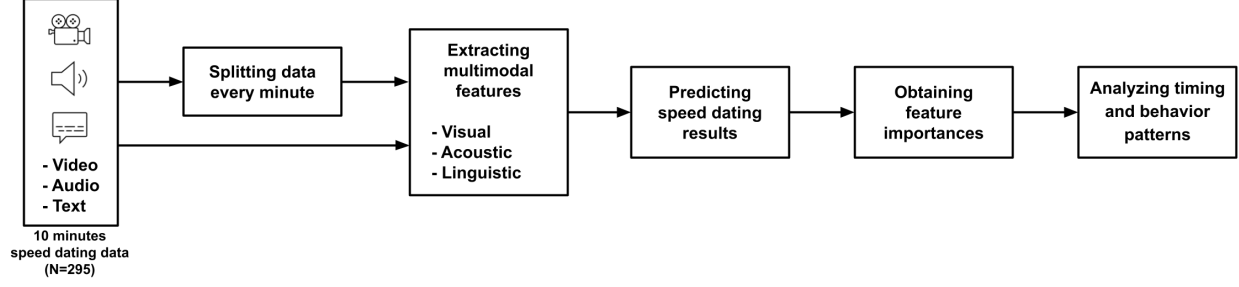


Figure 1: Workflow of this study

4.1 Feature Extraction

Existing studies have focused on analyzing only one or two features, such as acoustic or linguistic features [17, 23, 24, 29], or solely visual features [1]. However, to identify specific behavioral patterns at critical moments that influence outcomes of speed dating, it is essential to comprehensively analyze these features of participants' behaviors. Therefore, this study adopts a comprehensive approach by integrating visual, acoustic, and linguistic behavioral features previously utilized in related studies. To predict the outcomes of speed dating sessions based on participants' behaviors, we extracted visual, acoustic, and linguistic features related to participants' behaviors during the sessions from the video data, audio data, and transcribed textual data in the MMSD corpus. For visual features, we calculated statistical metrics related to head movements, gaze behavior, and Action Units. For acoustic features, we used statistical metrics of intensity, loudness, 12 Mel-Frequency Cepstrum Coefficients (MFCCs), pitch, probability of voicing, envelope, 8 LSF (Line Spectral Frequencies), and zero-crossing rate. For linguistic features, we utilized high-dimensional vectors extracted by natural language processing models.

4.1.1 Visual Features. Using the facial image processing tool OpenFace [3], we calculated statistical metrics related to head movements, gaze, and Action Units from the video data of the speed dating participants. For head movement features, we used the variance, median, 10th percentile, and 90th percentile values of the rotation angles along the x-, y-, and z-axes. In the video data, which captures the participants from a frontal view, the x-axis corresponds to the left-to-right direction from the camera's perspective, the y-axis corresponds to the bottom-to-top direction, and the z-axis corresponds to the front-to-back direction. For gaze behavior features, we used the variance, median, 10th percentile, and 90th percentile values of the gaze angles along the x- and y-axes. In the frontal-view video data, the x-axis represents the left-to-right direction, and the y-axis represents the bottom-to-top direction from the camera's perspective. For Action Unit (AU) features, we used the variance, median, 10th percentile, and 90th percentile values of the intensity of 17 Action Units. The Action Units used as features are listed in Table 1. As a result, the visual features amounted to a total of 88 dimensions.

4.1.2 Acoustic Features. Using the audio processing tool openSMILE [6], we extracted statistical metrics related to intensity, loudness, 12 MFCCs, pitch, probability of voicing, envelope, 8 LSFs, and zero-crossing rate from the audio data of the speed dating participants. The statistical measures calculated from the acoustic features are detailed in Tables 2 and 3, respectively. These features are provided as part of the standard feature set [25]. As a result, the acoustic features amounted to a total of 336 dimensions.

Table 1: List of Action Units.

Item	Content
AU01	Inner brow raiser
AU02	Outer brow raiser
AU04	Brow lowerer
AU05	Upper lid raiser
AU06	Cheek raiser
AU07	Lid tightener
AU09	Nose wrinkler
AU10	Upper lip raiser
AU12	Lip corner puller
AU14	Dimpler
AU15	Lip corner depressor
AU17	Chin raiser
AU20	Lip stretcher
AU23	Lip tightener
AU25	Lips part
AU26	Jaw drop
AU45	Blink

4.1.3 Linguistic Features. Using the natural language processing model BERT [5], we extracted 768-dimensional vectors from the transcribed textual data of the participants' speech during the speed dating sessions. For this purpose, we employed a Japanese pre-trained BERT model [11]. As a result, the linguistic features amounted to a total of 768 dimensions.

4.2 Experiment

To predict the outcomes of speed dating sessions based on participants' behaviors, we constructed a machine learning model using

Table 2: Statistics for acoustic features.

Statistic	Description
max	Maximum value
min	Minimum value
range	Difference between maximum and minimum
maxPos	Absolute position of maximum value
minPos	Absolute position of minimum value
amean	Mean value
stddev	Standard deviation
linregc1	Slope of linear approximation
linregc2	Offset of linear approximation
linregerrA	Difference between linear approximation and actual value
linregerrQ	Squared error of linear approximation
skewness	Skewness value
kurtosis	Kurtosis value

Table 3: Details of acoustic features.

Feature	Description
pcm_intensity_sma	Normalized intensity value
pcm_loudness_sma	Intensity value raised to 0.3 after normalization
mfcc_sma[1]–[12]	MFCC for 1–12
lspFreq_sma[0]–[7]	Frequencies calculated from 8 LPC coefficients
pcm_zcr_sma	Zero crossing rate
voiceProb_sma	Probability of being voice
F0_sma	Fundamental frequency
F0env_sma	Envelope of fundamental frequency

Table 4: Hyperparameter search space.

Hyperparameter	Search space
Number of trees	10, 20, 30, 40, 50, 60, 70, 80, 90, 100
Tree maximum depth	3, 4, 5, 6, 7, 8, 9, 10

Random Forest. Although the MMSD corpus contains data from a total of 625 speed dating sessions, this study used only the sessions where both participants mutually agreed to exchange contact information or mutually decided not to exchange it. Consequently, the dataset used in this study comprised 295 sessions, including 80 labeled as “want to exchange contact information” and 215 as “do not want to exchange contact information.” To gain insights into which time segments of behavior are important, the data from each speed dating was divided into 10 one-minute segments. For each segment, the features described in Section 4.1 were extracted, and these features were combined to form the input to the classification model. The output of the classification model was a binary value representing the outcome of the speed dating, indicating whether both participants mutually agreed to exchange contact information or not. Given the possibility that the influence of participants’ behaviors on the outcome might vary depending on the timing during the speed dating, we also built classification models to predict the outcome for each of the 10 one-minute segments. For evaluation, we employed a 10-fold cross-validation method. The number of trees

and tree maximum depth of the Random Forest model, which are key hyperparameters, were optimized through grid search within the search ranges specified in Table 4. As a baseline, for each trial in the 10-fold cross-validation, we created a model that randomly predicted class outputs based on the class distribution of the training data.

4.3 Result

Hereafter, a model that uses data from $i-1$ minutes to i minutes is referred to as M_i . For example, a model using data from 0–1 minute is denoted as M_1 . Performance metrics across models are listed in Table 5. An independent t-test on the F1-values, with Holm correction applied, was conducted between the baseline model and each model. The results of the t-test showed a significant difference at the 5% level between the baseline model and M_3 , and a marginally significant trend at the 10% level between the baseline model and M_4 , as well as the baseline model and M_{10} . No significant differences or trends were found between the baseline model and other models. The top 20 most important features in the constructed models were aggregated by modality. The ratios of features in the top 1% of feature importance for each model, categorized by modality, are shown in Table 6.

5 Discussion

The experimental results suggested that the factors influencing the outcomes of speed dating are: (1) behaviors occurring 2–4 minutes and 9–10 minutes after the start of the session, and (2) among these behaviors, the linguistic and acoustic elements. (1) Behaviors occurring 2–4 minutes and 9–10 minutes after the start of the session were used to train M_3 , M_4 , and M_{10} , contributing to the high performance of these models. This indicates a strong relationship between behaviors in these time segments and the outcomes of speed dating. This trend aligns with existing findings that report the importance of behaviors during the first and last quarters of the session [1]. Moreover, our results suggest that behaviors during the first two minutes, which did not contribute to the high performance of M_1 and M_2 , may have little impact on the outcomes. This provides new insights, indicating that behaviors in the very early

Table 5: Performance metrics across models.

Model	Segment	Precision	Recall	F1-value
Baseline	-	0.457	0.466	0.455
M_{ALL}	0-10 min	0.656	0.609	0.595
M_1	0-1 min	0.560	0.603	0.585
M_2	1-2 min	0.574	0.583	0.560
M_3	2-3 min	0.658	0.650	0.642
M_4	3-4 min	0.608	0.628	0.602
M_5	4-5 min	0.604	0.602	0.596
M_6	5-6 min	0.573	0.567	0.554
M_7	6-7 min	0.597	0.589	0.574
M_8	7-8 min	0.708	0.595	0.575
M_9	8-9 min	0.590	0.598	0.585
M_{10}	9-10 min	0.612	0.604	0.600

Table 6: Ratio of modalities to which the feature belongs among the top 1% of important features

Model	Visual	Acoustic	Linguistic
M_{ALL}	0.084	0.265	0.651
M_1	0.043	0.217	0.739
M_2	0.087	0.391	0.522
M_3	0.043	0.261	0.696
M_4	0.087	0.391	0.522
M_5	0.000	0.522	0.478
M_6	0.174	0.174	0.652
M_7	0.087	0.304	0.609
M_8	0.043	0.217	0.739
M_9	0.043	0.391	0.565
M_{10}	0.000	0.217	0.783

moments of the first quarter may be less critical. (2) The linguistic and acoustic-related features accounted for a large proportion of the important features in M_3 , M_4 , and M_{10} . Conversely, visual features contributed little to the performance of these models, suggesting that they may have limited influence on the outcomes of speed dating. This represents a novel finding uncovered through the multimodal analysis conducted in this study.

6 Conclusion

We explored the timing and behavior patterns affecting speed dating outcomes. Specifically, we developed machine learning models to predict outcomes using multimodal features from each segment and analyzed the relationship between prediction performance and the segments and features used. From the performance metrics of the constructed models, it was suggested that using participants' behaviors during the 2–4 minute and 9–10 minute intervals of a 10-minute speed dating could predict outcomes as effectively as, or even better than, using data from the entire speed dating duration. Furthermore, among the visual, acoustic, and linguistic behaviors, linguistic behaviors were suggested to be the most critical for predicting the outcomes of speed dating. Future research should investigate which specific behaviors within each modality

influence the prediction of speed-dating outcomes. Improvements to feature extraction could include treating the data as sequential to capture context and incorporating features that represent the interaction between participants. Although the above findings were obtained, this study has three significant limitations: first, the participants were all Japanese; second, the speed dating sessions were limited to 10 minutes; and third, only data from cases where both participants agreed on the matching outcome were analyzed. It would be highly interesting to examine whether the findings of this study hold true when participants are from different nationalities or when the duration of the speed dating differs. Additionally, while this study focused on cases where both participants' matching outcomes aligned as an initial investigation, future research should also analyze the behaviors and timing of participants in cases where their matching outcomes differ, to gain a deeper understanding of the factors influencing matching results.

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